MIL-C-48244 (PA)
3 December 1973
SUPERSESSION
(See 6.4)

MILITARY SPECIFICATION

CELLULOSE, POWDERED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of purified wood cellulose for use in the formulation of ammunition inert fillers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber.
RR-S-366 - Sieves: Standard, Testing.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The material shall consist of purified wood cellulose which conforms to the following requirements:

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Property	Min	Max	Applicable Paragraph
Moisture, %	5.0	7.0	4.4.1
Screen size on			
35mesh, %	---	1.0	4.4.2
thru 100mesh,%	96	---	

3.2 Foriegn Material.-The composition shall be free of dirt, chips, grease and other foreign matter when examined as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define the terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and quality assurance provisions specified in this specification.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications, shall be made available upon request by the contracting officer.

d. Quantity of product in the lot.

e. Date submitted.

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The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to initiation of regular production the contractor shall submit a first article sample consisting of 200 grams of powdered cellulose in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

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4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of one or more batches of powdered cellulose produced by one manufacturer in accordance with the same specification, or same specification revision under one continuous set of operating conditions. Each batch shall consist of that quantity of the powdered cellulose that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). The criteria and procedures for assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following specification of defects shall be in accordance with MIL-STD-105. Continuous sampling plans, in accordance with MIL-STD-1235, may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.3.2.1 Drums, prior to filling (Polyethylene Liner).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Liner cut, torn, or punctured	Visual	01001
Minor:	None defined.		

4.3.2.2 Drums, prior to closing (Polyethylene liner).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Liner not completely sealed	Visual	02001
Minor:	None defined.		

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4.3.2.3 Drum, sealed (fiber).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Weight of contents	Scale	03001
102.	Closure incomplete or damaged to the extent that contents sift out	Visual	03002
Minor:	AQL 0.65 percent		
201.	Markings misleading or unidentifiable	Visual	03003

4.3.3 Sampling.-Approximately 5 pounds of powdered cellulose shall be withdrawn from each batch to be sampled using ASTM Procedure E300-70 for solids. Sample shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.4. If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification and code numbers for the tests are given below:

Category	Defect	Code No. (see 6.2)
Moisture (see 3.1)	Major	04001
Screen Size (see 3.1)	Major	05001
Foreign Material (see 3.2)	Major	06001

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Moisture.-Transfer quantitatively a 2.0 gram sample of the flock to a tared container at least two inches in diameter. Dry the container and contents in an oven maintained at 105°C ± 5°C for a minimum of four hours. Remove container and contents from the oven and transfer to a desdicator until cool and then weigh.

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$$\text{Percent Moisture} = \frac{(B-A)\ X100}{W}$$

Where: B = Weight of container and sample before drying.
A = Weight of container and sample after drying.
W = Weight of sample.

4.4.1.2 Moisture (Alternate Method).-Use of an on-line instrumental method for the determination of moisture content may be authorized with the approval of the Contracting Officer (see 6.5).

4.4.2 Screen size.-Determine the screen size of the sample using 35 and 100 mesh U.S. Standard Sieves (U.S.S.S.) conforming to Federal Specification RR-S-366 as follows:

Place 100 $\pm$ 0.1 gm of the sample on the top sieve of the specified nest of sieves properly superimposed and assembled with a bottom pan. Cover the assembly and shake for 5 minutes by hand or by means of a mechanically operated sieve shaker, which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 $\pm$ 15 gyrations and 150 $\pm$ 10 taps of the striker per minute. The portions retained or passed by the various sieves shall be weighed and the results calculated to a percentage bases as required.

4.4.3 Foreign Material.-Spread the sample of material obtained in 4.3.3 on a sheet of Kraft paper (approximate 3 feet by 5 feet) and visually examine for presence of Foreign Material.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.-The powdered cellulose shall be packed in fiber drums complying with Type III Grade A of Specification PPP-D-723. The drums shall be finished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure.

5.1.2 Level C.-The packing shall be as stated in the contract of order.

5.2 Marking.-In addition to any special markings required by the contract or purchase order, containers shall be marked in accordance with MIL-STD-129. Marking shall include, but not limited to, the following information:

a. Manufacturers name.
b. Product designation.
c. Lot number.
d. Date of manufacture.
e. Number of this specification.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify The following:

a. The title, number, and date of this document.

b. Acceptance and Description sheets.-These sheets shall be prepared for each lot in accordance with MIL-STD-1171.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use.-The cellulose covered in this specification is intended to be used as an ingredient in the manufacture of inert fillers for ammunition.

6.4 Supersession Data.-This specification includes the requirements of Purchase Description PA-PD-2765, dated 10 February 1966.

6.5 Approval of alternate method.-Completion details of proposed method should be submitted to Picatinny Arsenal, Dover, New Jersey 07801, Attn: SARPA-QA-A-P thru the Contracting Officer for review and approval.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project Number 1395-A004

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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